

GUMPOLDSKIRCHEN, Austria

WMO: 110820

Lat: 48.0358N

Lon: 16.2808E

Elev: 696

StdP: 14.33

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	19.0	7.0	8.1	17.5	11.7	10.2	22.0	24.1	37.8	20.4	39.7	3.5	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	89.4	70.6	85.9	69.2	82.8	67.6	71.9	86.0	70.3	83.5	68.7	80.6	5.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	102.7	78.8	65.6	96.8	76.8	64.1	91.9	75.8	36.1	86.1	34.6	83.9	33.3	80.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	16.2	13.7	10.5	94.2	5.3	4.3	6.7	97.3	3.6	99.8	0.6	102.2	-3.3	105.3
			9.3	74.5	5.1	1.6	5.6	75.7	2.7	76.6	-0.2	77.6	-3.9	78.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	32.9	36.5	43.6	52.9	60.6	67.2	70.7	70.0	61.1	51.8	43.1	34.8	(6)
(7)		DBStd	15.03	8.21	8.53	7.75	7.34	6.38	6.35	5.85	5.72	6.03	6.94	7.18	7.75	(7)
(8)		HDD50	1960	531	382	224	55	3	0	0	2	63	227	473	(8)	
(9)		HDD65	5222	994	797	664	366	165	50	16	17	148	411	658	936	(9)
(10)		CDD50	2754	2	5	26	142	330	516	641	620	333	119	19	1	(10)
(11)		CDD65	542	0	0	0	3	28	116	193	172	29	1	0	0	(11)
(12)		CDH74	4359	0	0	0	35	218	922	1597	1401	178	8	0	0	(12)
(13)		CDH80	1353	0	0	0	4	28	257	545	488	31	0	0	0	(13)
(14)	Wind	WSAvg	5.5	5.6	5.6	5.7	6.0	6.2	5.9	6.2	5.4	5.3	4.5	4.8	5.1	(14)
(15)	Precipitation	PrecAvg	25.6	1.4	1.3	1.9	1.7	2.8	3.1	3.1	2.8	2.6	1.8	1.7	1.5	(15)
(16)		PrecMax	34.1	3.4	2.9	4.7	3.9	6.6	7.0	8.9	6.7	7.9	3.8	3.4	2.7	(16)
(17)		PrecMin	17.0	0.1	0.0	0.6	0.0	0.7	0.9	0.3	0.6	0.8	0.2	0.0	0.4	(17)
(18)		PrecStd	4.4	0.8	0.9	1.1	1.1	1.4	1.5	1.7	1.6	1.6	0.9	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	61.7	68.3	78.2	83.3	90.5	93.9	94.9	83.9	74.1	65.4	57.0	(19)
(20)		MCWB	48.2	48.7	53.5	60.8	66.9	70.6	72.0	70.2	66.5	61.5	55.4	49.1	(20)	
(21)		2%	DB	53.1	56.5	63.5	73.1	79.4	86.3	89.4	89.5	79.0	69.3	61.1	52.9	(21)
(22)		MCWB	46.0	47.3	51.3	58.4	64.7	69.7	70.8	70.4	64.9	59.3	53.3	45.9	(22)	
(23)		5%	DB	48.8	52.6	59.3	69.1	76.2	82.7	85.8	85.4	74.8	65.8	56.8	48.6	(23)
(24)		MCWB	43.1	45.1	49.1	56.2	63.0	67.8	69.1	69.1	62.9	57.6	50.7	43.2	(24)	
(25)		10%	DB	44.9	48.4	55.5	65.5	72.7	79.4	82.5	81.6	71.5	62.5	53.2	45.1	(25)
(26)		MCWB	40.2	42.7	47.3	54.2	60.8	66.0	67.7	67.1	61.3	55.8	48.4	40.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	50.3	54.7	62.0	68.4	73.8	74.6	73.6	68.1	62.7	56.7	49.5	(27)
(28)		MCD B	56.3	59.2	65.6	75.9	80.7	86.4	89.1	89.1	81.2	71.0	63.2	55.3	(28)	
(29)		2%	WB	46.4	47.6	52.0	59.2	65.9	70.9	72.0	71.5	66.0	60.5	53.9	46.4	(29)
(30)		MCD B	52.3	55.6	61.5	71.9	77.7	83.7	86.2	85.9	76.9	67.4	60.0	52.2	(30)	
(31)		5%	WB	43.3	45.5	50.1	56.9	63.8	68.9	70.3	69.8	64.1	58.5	51.3	43.5	(31)
(32)		MCD B	48.1	52.1	58.1	67.4	74.1	80.7	83.4	83.2	72.9	64.4	56.3	48.1	(32)	
(33)		10%	WB	40.5	43.0	48.1	55.0	61.9	67.1	68.7	68.0	62.3	56.6	48.9	41.1	(33)
(34)		MCD B	44.5	48.4	54.7	64.1	71.2	77.7	80.3	79.7	69.8	61.6	52.7	44.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	8.9	11.2	13.7	16.4	16.2	16.4	16.7	17.5	14.6	13.6	9.5	8.6	(35)
(36)		5% DB	MCD BR	14.6	16.9	21.2	23.3	22.8	22.7	23.2	24.6	21.2	18.6	15.0	14.9	(36)
(37)		MCWBR	9.6	10.6	12.0	11.8	11.0	9.7	8.7	9.2	9.5	10.2	9.3	10.0	(37)	
(38)		5% WB	MCD BR	13.9	15.8	19.9	21.7	21.5	21.5	21.4	22.8	19.9	17.3	14.0	13.8	(38)
(39)		MCWBR	9.7	10.4	11.7	11.2	10.6	9.7	8.5	9.1	9.3	10.0	9.3	9.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.342	0.382	0.427	0.426	0.431	0.437	0.437	0.408	0.377	0.347	0.316	(40)	
(41)		taud	2.385	2.341	2.261	2.167	2.217	2.241	2.226	2.241	2.290	2.349	2.386	2.413	(41)	
(42)		Ebn at Noon	238	256	262	261	266	265	261	255	251	239	222	223	(42)	
(43)		Edh at Noon	22	29	36	44	43	43	43	40	35	28	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	323	585	952	1409	1658	1807	1752	1562	1118	693	361	254	(44)	
(45)		RadStd	38	78	121	144	177	136	158	154	136	76	37	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)	
		+1.42	N/A	N/A	+2.14	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (91 neighbors)		+1.01	N/A	N/A	+1.51	+0.81	+0.68	-216	-288	+153	+77	(47)

Nomenclature: See separate page